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the memory automatically. A user cannot communicate directly with the computer hardware, unless an operating system is loaded.

Some of the popular Operating Systems used in personal computers and laptops are **Windows**, **UNIX** and **Linux**. The mobile devices mostly use Android and ioS as mobile **OS.**

4.2.1 Need for Operating System

Operating System has become essential to enable the users to design applications without the knowledge of the computer's internal structure of hardware. Operating System manages all the Software and Hardware. Most of the time there are many different computer programmes running at the same time, they all need to access the Computers, CPU, Memory and Storage. The need of Operating System is basically - an interface between the user and hardware.

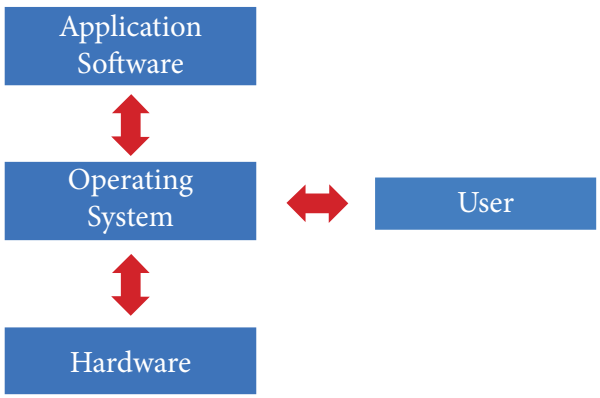

Figure: 4.2 Interaction of Operating system and user

Operating System works as translator, while it translates the user request into machine language(Binary language), processes it and then sends it back to Operating System. Operating System converts processed information into user readable form

Uses of Operating Systems

The following are few uses of Operating System

The main use of Operating System is

- to ensure that a computer can be used to extract what the user wants it do.
- Easy interaction between the users and computers.
- Starting computer operation automatically when power is turned on (Booting).
- Controlling Input and Output Devices
- Manage the utilization of main memory.
- Providing security to user programs.

4.3 Types of Operating System

Operating System are classified into the following types depending on their processing capabilities.

4.3.1 Single User Operating Systems

An operating system allows only a single user to perform a task at a time. It is called as a Single user and single Task operating system. For a user, a task is a function such as printing a document, writing a file to disk, editing a file or